

CAMPULUNG MUSCEL, RomaniaWMO: **153240**Lat: **45.2747N**Lon: **25.0367E**Elev: **2238**StdP: **13.55**Time Zone: **2.00 (EUE)**Period: **86-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	11.2	14.5	4.6	7.6	16.9	8.0	9.0	19.5	22.4	26.8	18.0	26.3	2.6	50	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	20.7	83.2	67.2	80.8	65.9	78.4	64.9	69.0	80.4	67.5	77.8	66.1	76.3	3.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
65.2	101.2	74.3	63.7	95.9	72.9	62.3	91.4	71.4	34.6	80.8	33.2	77.7	32.1	76.2	80.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 16.0	11.8	9.1	DB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			WB	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	48.4	29.2	34.3	39.2	47.3	56.4	63.8	65.3	66.0	57.6	48.9	39.6	32.0
	DBStd	14.58	7.83	7.74	7.93	7.78	5.52	5.26	4.79	4.98	6.41	7.76	7.82	7.61
	HDD50	2585	644	439	343	140	14	0	0	0	12	114	321	558
	HDD65	6270	1109	858	799	532	269	83	56	49	232	499	761	1023
	CDD50	1995	0	0	9	58	213	413	475	497	239	81	10	0
	CDD65	204	0	0	0	0	2	46	66	81	9	0	0	0
	CDH74	1855	0	0	0	5	45	415	575	705	105	5	0	0
	CDH80	304	0	0	0	0	1	68	97	132	6	0	0	0
Wind	WSAvg	2.7	2.4	2.6	3.4	3.2	2.9	2.8	2.8	2.6	2.6	2.3	2.3	2.3
Precipitation	PrecAvg	33.6	1.6	1.5	1.9	2.8	4.0	4.5	4.4	3.5	2.8	2.5	2.0	2.0
	PrecMax	45.9	3.0	2.9	3.3	5.4	7.0	8.0	7.8	7.3	6.4	5.6	4.8	4.1
	PrecMin	21.0	0.3	0.3	0.4	1.2	1.8	1.8	1.9	1.1	0.5	0.1	0.2	0.3
	PrecStd	6.5	0.8	0.7	0.9	1.1	1.6	1.7	1.7	1.7	1.5	1.3	1.1	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	54.1	59.6	67.5	74.0	78.4	86.6	86.3	86.8	80.7	74.5	64.4	56.5
		MCWB	42.6	48.9	51.3	56.7	62.7	69.6	70.4	69.0	62.2	58.7	51.5	44.8
	2%	DB	48.5	54.3	62.8	69.7	74.7	81.8	82.6	83.1	76.9	70.5	58.9	49.4
		MCWB	39.9	44.8	49.2	55.3	60.8	67.2	68.4	66.1	60.9	56.0	50.4	42.2
	5%	DB	43.4	49.7	57.3	65.3	71.5	78.7	79.9	80.8	72.9	66.3	54.9	45.8
		MCWB	37.8	41.7	46.4	52.0	59.4	66.0	66.5	65.2	59.8	55.1	48.6	41.0
	10%	DB	39.3	45.9	52.4	60.8	68.5	75.9	77.1	77.9	69.6	61.3	50.2	41.7
		MCWB	35.6	39.4	43.5	50.2	57.6	64.6	65.1	64.7	58.4	52.8	45.9	38.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	44.2	49.5	53.3	58.3	64.3	72.0	71.0	70.4	63.8	60.9	53.4	46.8
		MCDB	51.7	59.3	66.1	71.3	75.9	84.8	83.3	83.2	76.1	70.0	61.1	50.6
	2%	WB	40.9	45.6	50.2	55.9	62.0	68.6	69.0	68.4	62.4	57.8	51.6	44.0
		MCDB	46.5	53.0	62.2	67.8	72.9	79.6	79.9	79.3	73.9	66.6	57.7	48.8
	5%	WB	38.2	42.8	47.1	53.7	60.0	66.9	67.4	66.8	61.0	55.8	49.4	41.0
		MCDB	42.1	48.8	56.7	63.5	69.8	77.2	77.3	77.8	71.6	63.9	54.4	44.7
	10%	WB	35.8	40.5	44.4	51.4	58.2	65.4	65.9	65.4	59.6	53.9	46.8	38.6
		MCDB	38.5	45.2	51.8	59.7	67.1	75.0	75.0	76.2	68.5	59.8	49.7	41.8
Mean Daily Temperature Range	5% DB	MDBR	14.0	16.1	17.3	18.3	18.4	19.8	20.2	20.7	19.4	18.2	14.3	13.0
		MCDBR	19.8	22.4	25.6	25.4	23.6	23.8	23.3	24.4	23.5	25.5	20.8	18.0
		MCWBR	12.4	14.2	14.8	13.2	12.1	12.1	11.7	11.0	10.8	13.4	12.1	11.6
	5% WB	MCDBR	17.9	21.0	24.5	23.4	22.3	22.6	21.1	22.3	22.3	22.5	18.5	16.2
		MCWBR	11.3	13.6	14.3	12.5	12.0	12.3	11.3	10.7	10.3	12.4	11.0	10.7
Clear-Sky Solar Irradiance	taub		0.304	0.336	0.379	0.440	0.419	0.420	0.434	0.451	0.404	0.366	0.339	0.308
	taud		2.323	2.241	2.197	2.073	2.204	2.239	2.206	2.150	2.254	2.354	2.374	2.364
	Ebn at Noon		252	264	267	259	270	270	264	253	256	251	237	239
	Edh at Noon		25	33	40	49	44	43	44	45	37	29	24	22
All-Sky Solar Radiation	RadAvg		445	706	1055	1387	1692	1837	1865	1679	1229	840	487	364
	RadStd		50	94	112	150	158	159	152	133	115	107	52	46